

The Interdisciplinary Lens: Viewing Society Through Multiple Social Science Fields

Throughout this quarter, you have examined individual social science disciplines: psychology, counseling, social work, education, criminology, communication, multimedia arts, and gender studies. Each discipline offers valuable insights. Psychology reveals how individuals think and behave. Sociology exposes social structures and inequalities. Political science analyzes power and governance. Economics explains the production and distribution of resources. But real-world problems do not arrive neatly packaged as "a psychology problem" or "an economics problem." Poverty, climate change, pandemics, armed conflict, and inequality are complex. They have multiple causes, affect multiple groups, and require multiple types of expertise to address. This is why transdisciplinarity matters.

This reading material examines the value of transdisciplinarity in understanding society. It defines transdisciplinarity and distinguishes it from multidisciplinary and interdisciplinary. It explains why no single discipline is sufficient to address complex social problems. Finally, it demonstrates how integrating knowledge across and beyond disciplines produces richer, more useful understanding.

Part 1: What Is Transdisciplinarity?

Transdisciplinarity is the highest level of integration among disciplines. It goes beyond simply placing disciplines side by side (multidisciplinary) or even crossing boundaries between them (interdisciplinary). Transdisciplinarity integrates knowledge across academic disciplines and also reaches beyond the academy to include non-academic knowledge – the expertise of practitioners, policymakers, community members, and those who experience problems firsthand.

Three Levels of Integration

Level	What It Means	Relationship Among Disciplines	Includes Non-Academics?
Multidisciplinarity	Different disciplines study the same problem separately	Disciplines work in parallel, not together	No
Interdisciplinarity	Disciplines collaborate, integrating methods and theories	Disciplines work together, crossing boundaries	No
Transdisciplinarity	Disciplines integrate with each other and with non-academic knowledge	Deep integration across disciplines and beyond the academy	Yes

Why Transdisciplinarity Is Necessary

No single social science discipline can fully explain a complex social problem. Psychology focuses on individual behavior but may miss structural inequality. Sociology focuses on social structures but may overlook individual agency. Economics focuses on markets but may ignore cultural values. Political science focuses on power but may neglect psychological factors. Each discipline sees only a slice of reality. Transdisciplinarity assembles the whole picture.

Part 2: The Limitations of Single-Discipline Approaches

Consider a complex social problem: why do some communities have persistently high rates of poverty despite government intervention?

Discipline	What It Sees	What It Misses
Psychology	Individual factors: low self-esteem, poor decision-making, trauma, learned helplessness	Structural barriers (lack of jobs, discrimination, poor schools)
Sociology	Social structures: class inequality, residential segregation, weak social networks, lack of collective efficacy	Individual agency, psychological barriers, historical context
Economics	Market failures: lack of jobs, low wages, insufficient human capital, credit constraints	Cultural values, social networks, political power, historical legacies
Political science	Policy failures: poorly designed programs, corruption, lack of political will	Lived experience of poverty, psychological impacts, community knowledge
History	Historical legacies: colonial land distribution, past discriminatory policies, path dependence	Current conditions, individual experiences, immediate barriers

Each discipline is correct, but each is incomplete. Poverty is not merely a psychological problem (fix the person) nor merely an economic problem (fix the market) nor merely a political problem (fix the policy). It is all of these simultaneously. Understanding poverty requires integrating insights from multiple disciplines – and also listening to people who experience poverty themselves.

Part 3: The Value of Transdisciplinarity

Transdisciplinarity is not merely an academic exercise for scholars who enjoy using complicated jargon. Transdisciplinarity matters because real-world problems resist single-discipline solutions. A psychologist studying poverty might focus on individual decision-making and trauma. An economist might focus on labor markets and credit

constraints. A sociologist might focus on class structures and social networks. Each perspective is valuable, but each is incomplete. The person experiencing poverty does not experience it as a psychological problem or an economic problem or a sociological problem. They experience it as all of these simultaneously. Transdisciplinarity responds to this reality. It brings multiple disciplines into conversation, corrects their blind spots, and generates insights that no single discipline could produce alone. The table below summarizes five key values of transdisciplinarity, with examples drawn from social science research.

What Transdisciplinarity Adds

Value	Explanation	Example
Completeness	Multiple perspectives reveal dimensions a single approach would miss	Understanding poverty requires psychology (individual), sociology (structural), economics (market), political science (policy), history (legacy)
Mutual correction	Insights from one discipline can challenge false assumptions in another	Economics assumes rational actors; psychology shows systematic biases in decision-making
Innovation	Creative solutions often emerge at the boundaries between disciplines	Behavioral economics (psychology + economics) has produced new anti-poverty interventions
Relevance	Including non-academic knowledge ensures research addresses real needs	People who experience poverty know barriers that researchers might not anticipate
Problem-focused	Instead of being organized by discipline, research is organized by the problem itself	A transdisciplinary poverty team starts with the problem, not with psychology or economics

Examples of Transdisciplinary Success

Example 1: Reducing Child Malnutrition

Discipline	Contribution
Nutrition science	Identified what nutrients children need
Economics	Analyzed why poor families cannot afford nutritious food
Psychology	Studied how mothers make feeding decisions; addressed beliefs and habits
Sociology	Examined community norms around child feeding; social support networks
Political science	Studied why nutrition programs are underfunded or poorly implemented
Anthropology	Understood cultural meanings of food and feeding practices
Public health	Designed and evaluated interventions

Transdisciplinary addition: Including mothers (non-academic experts) revealed that they already knew about nutritious food but lacked time, money, and household decision-making power. The solution was not more nutrition education but addressing women's workload and economic empowerment.

Example 2: Responding to the COVID-19 Pandemic

Discipline	Contribution
Epidemiology	Tracked disease spread; identified risk factors
Virology	Studied virus transmission
Psychology	Studied why people refused masks or vaccines; developed effective messaging
Sociology	Analyzed how lockdowns affected different social classes differently

Discipline	Contribution
Economics	Estimated economic impacts; designed relief programs
Political science	Compared government responses across countries
Communication	Studied how misinformation spread; designed public health campaigns
Ethics	Weighed competing values: health vs. economy, liberty vs. safety

Transdisciplinary addition: Including healthcare workers (practical knowledge), educators (school closures), small business owners (economic realities), and community leaders (trust messaging) ensured policies were feasible and accepted.

Part 4: Transdisciplinarity in Practice – When and How

When to Use a Transdisciplinary Approach

Situation	Why Transdisciplinarity Helps
The problem has multiple causes	Captures the full range of causal factors that no single discipline can cover alone
The problem affects multiple groups differently	Ensures solutions address differential impacts on gender, class, race, age, and location
Single-discipline solutions have failed	Brings in overlooked disciplines to identify what previous approaches missed
Stakeholders disagree on causes and solutions	Creates a shared framework that reveals hidden assumptions and common ground

Situation

Why Transdisciplinarity Helps

The problem requires action, not just understanding

Includes practitioners and community members who know what is feasible

Challenges of Transdisciplinary Work

Challenge

How to Address It

Different jargon

Invest time in building shared vocabulary

Power imbalances among disciplines

Actively value diverse forms of expertise

Time and resources

Plan for extended timelines; secure adequate funding

Institutional barriers (disciplines are separated in universities)

Develop transdisciplinary programs, centers, and funding streams

Evaluating quality

Develop appropriate criteria: real-world impact, stakeholder satisfaction, problem-solving effectiveness

Key Terms

Term	Definition
Transdisciplinarity	Integration of knowledge across academic disciplines and beyond the academy (practitioners, community members, policymakers)
Multidisciplinarity	Different disciplines study the same problem separately, without integration
Interdisciplinarity	Different disciplines collaborate, integrating methods and theories
Complex problem	A problem with multiple interacting causes, no single solution, and no clear stopping point (e.g., poverty, climate change)
Mutual correction	When insights from one discipline challenge false assumptions in another
Problem-focused	Organizing research around the problem itself, not around disciplinary boundaries

Review Questions

1. Define transdisciplinarity. How does it differ from multidisciplinary and interdisciplinarity?

2. Complete the following table:

Level	Definition	Includes Non-Academics?
Multidisciplinary		
Interdisciplinarity		
Transdisciplinarity		

3. Choose one complex social problem (e.g., poverty, climate change, crime, inequality, public health). Identify three different social science disciplines that would contribute to understanding this problem. For each discipline, state one thing it would contribute that other disciplines might miss.

4. What is "mutual correction" in transdisciplinary work? Provide one example of how one discipline might correct a false assumption in another discipline.

5. List two challenges of transdisciplinary work. For each challenge, describe one way to address it.

6. Why is it important to include non-academic knowledge (from practitioners, community members, people with lived experience) in transdisciplinary research? What do they know that academic researchers might miss?

7. Application question: A university is planning a research center on urban poverty. Using the concept of transdisciplinarity, describe what disciplines should be included, who from outside the university should be involved, and how the center would operate differently from a traditional single-discipline department.

8. Reflection question: Of the six Q2 lessons (Psychology and Counseling, Social Work, Education, Criminology, Communication and Multimedia, Gender Studies), which two do you think are most important for understanding poverty? Why?